

ArchViz Pro

3D Office Floor Planner & Asset Manager

User Manual

An immersive CAD digital-twin tool for planning office layouts and low-current network / AV infrastructure.

Version 1.0

Contents

1. Introduction
2. Getting Started
3. The Interface at a Glance
4. Navigating the 3D Workspace
5. Working with Assets
6. Working with Rooms
7. Analytics Overlays
8. Using Blueprints
9. AI Features
10. Inventory, Reports & Exports
11. Saving Your Work
12. Keyboard & Mouse Reference
13. Troubleshooting & FAQ

1. Introduction

ArchViz Pro is a browser-based **digital-twin designer** for offices. It combines an interactive 3D viewport with a low-current asset manager, so facilities, IT and audio-visual teams can plan and audit a space in a single tool. You can lay out rooms, drop in furniture and networking devices, visualise Wi-Fi coverage, occupancy load and cabling paths — and even let AI turn a real blueprint into an editable 3D model.

Who is it for?

- **Architects & space planners** — lay out rooms and furniture.
- **IT & network teams** — place access points, data points, CCTV and cabling.
- **AV teams** — position HDMI / projector ports and interactive displays.
- **Facilities managers** — audit occupancy, coverage and inventory.

No installation needed. ArchViz Pro runs entirely in your web browser. Your design is saved automatically to that browser.

2. Getting Started

When you first open ArchViz Pro, a short onboarding guide appears. Read the three tips and click **Launch CAD Workspace** to begin. A sample office layout is loaded so you can start exploring immediately.

First steps

- 1 Use your mouse to orbit, pan and zoom around the 3D scene.
- 2 Click any device or piece of furniture to select it.
- 3 Open the **Library** tab on the right to add new items.
- 4 Switch between **3D View** and **2D Plan** using the toggle in the header.
- 5 When you are happy, use **Share & Export** to save a copy.

Re-open the guide any time. Click the ? (help) icon on the far-left rail to show the onboarding guide again.

3. The Interface at a Glance

The screen is divided into a few clear zones:

Area	Location	What it does
Header	Top bar	Project name, 2D/3D toggle, Heatmap, Reset, API Key, Share & Export.
Icon rail	Far left	Switches the side panel between Library, Rooms and Inventory; opens Help.
3D Workspace	Centre	The live, interactive 3D / 2D model of your office.
Side panel	Right	Add items, edit the selected asset, manage rooms, run reports.
Inspector HUD	Bottom-left	Live details when you hover an asset or room.
Footer	Bottom bar	Scale and grid-snap reference information.

4. Navigating the 3D Workspace

- **Orbit / rotate** — click and drag on empty space (3D view only).
- **Pan** — right-click and drag, to slide the view sideways.
- **Zoom** — scroll the mouse wheel or use a trackpad pinch.
- **2D Plan view** — locks the camera to a top-down blueprint angle, ideal for precise placement and printing.

2D is best for placement. Switch to **2D Plan** when you want to position items exactly on the floor grid without perspective distortion.

5. Working with Assets

Assets are the devices and furniture you place in the space. They fall into two categories: **Low-Current (infrastructure)** and **Furniture**.

Adding an asset

- 1 Open the **Library** tab in the side panel.
- 2 Optionally filter by **All**, **Low-Current** or **Furniture**.
- 3 Click an item card to add it to the centre of the floor.
- 4 The new item is selected automatically so you can edit its details.

Moving, rotating & editing

- **Move** — click and drag the item across the floor.
- **Select** — click once; a cyan ring highlights the selection.
- **Edit** — with an item selected, use the inspector to change its name, specifications, rotation and room assignment.
- **Delete** — use the delete (trash) control in the inspector.

Placement warnings

If an asset overlaps another item on the same height layer, or sits outside every room boundary, it turns **red** with a warning ring and beacon. The bottom-left inspector explains the exact issue. Ceiling-mounted devices (access points, CCTV) never collide with floor items.

Asset catalogue

Category	Items
Low-Current	Access Point (AP), Data / LAN Port (DP), Telephone Port (TP), Power Outlet, HDMI Port, Projector Port, CCTV Camera, Door Access, Intercom
Furniture	Single Desk, 4-Pax & 6-Pax Desk Clusters, Conference Table, Office Chair, Lounge Armchair, Reception Counter, Whiteboard

6. Working with Rooms

Rooms define the zones of your floor plan. Each room has a name, dimensions, colour and an automatically calculated area and occupancy capacity.

Creating and managing rooms

- 1 Open the **Rooms** tab in the side panel.
- 2 Enter a name, width, depth and colour, then add the room.
- 3 Assets dropped inside a room are automatically assigned to it.
- 4 Delete a room to free its assets (they become unassigned, not deleted).

Room setup templates

Save time with one-click templates that populate a room with a coordinated set of furniture and infrastructure:

- **Boardroom** — conference table, eight chairs, whiteboard, AP and AV ports.
- **4-Workstation cluster** — a 4-person desk cluster, chairs and a data point.
- **Executive suite** — desk, chairs, lounge seat, whiteboard, phone and CCTV.

7. Analytics Overlays

Overlays turn your model into a live analysis tool. Toggle them from the header or the Library tab.

Overlay	Shows	How to read it
Occupancy Heatmap	Room density vs. capacity	Green = comfortable, amber = busy, red = overloaded.
Wi-Fi Coverage	Distance to nearest access point	Mint = excellent, yellow = fair, rose = weak / dead zone.
Cabling Paths	Cable runs to the network room	Colour-coded by service (AP, data, voice, CCTV).

Occupancy uses 2D view. Turning on the Heatmap automatically switches to 2D Plan for the clearest reading.

8. Using Blueprints

You can trace over a real floor plan by loading it as a ground underlay.

- 1 Open the **Rooms** tab and use the blueprint uploader.
- 2 Choose an image (PNG / JPG) or a PDF — page 1 is rendered automatically.
- 3 Adjust **opacity**, **scale** and **offset** so it aligns with the grid.
- 4 Draw rooms and place assets on top, then hide the underlay when finished.

9. AI Features

ArchViz Pro can use Google Gemini to build a layout for you. AI features use a **Bring Your Own Key** model.

Setting your API key

- 1 Click **Set API Key** in the header.
- 2 Paste your Google Gemini key (free from Google AI Studio) and accept the terms.
- 3 The key is stored only in your browser session and is removed when you close the tab.

Digitize a blueprint

Upload a floor-plan image, then run the AI digitizer. Gemini detects rooms and suggests an initial layout of access points, data points, cameras and furniture — all fully editable afterwards.

Rebuild from a prompt

Describe the office in plain language — for example, *“open-plan startup floor with a boardroom, two meeting pods and a reception”* — and generate a complete layout to refine.

Your key stays private. Keys are never stored on the server — they are sent securely with each request and billed to your own Google account.

10. Inventory, Reports & Exports

The **Inventory** tab lists every asset with search and filtering, and lets you export the data. The **Share & Export** dialog (header) offers richer output formats.

Export	Where	Contents
PNG Snapshot	Share & Export	An image of the current 3D view.
PDF Report	Share & Export	Snapshot plus room summary and full asset inventory.
JSON (download)	Both	Re-importable schematic backup of the whole design.
Copy JSON	Share & Export	Copies the layout to your clipboard.
CSV	Inventory tab	Spreadsheet of assets, rooms and coordinates.

To save a PDF report, the report opens in a new browser tab — choose **Save as PDF** in the print dialog. Allow pop-ups if nothing appears.

11. Saving Your Work

- **Autosave** — your design is saved to this browser automatically as you work.
- **Reload-safe** — refreshing the page restores your latest layout.
- **Reset Workspace** — clears everything (and the saved copy); use with care.
- **Portable backup** — export JSON to move a design between browsers or devices.

Back up important designs. Because autosave is tied to one browser, export a JSON file for any design you want to keep long-term or share.

12. Keyboard & Mouse Reference

Action	Control
Orbit / rotate view (3D)	Left-click + drag on empty space
Pan view	Right-click + drag
Zoom in / out	Mouse wheel / trackpad pinch
Select an asset	Left-click the asset
Move an asset	Left-click + hold on the asset, then drag
Deselect	Left-click empty space
Switch 2D / 3D	Header toggle

13. Troubleshooting & FAQ

Issue	Solution
"Gemini API Key is required."	Click Set API Key and paste your Gemini key.

Issue	Solution
AI digitizer shows an error	Usually transient demand — wait a moment and retry.
PNG export is blank	Let the 3D view finish rendering, then export again.
PDF report won't open	Allow pop-ups for the site; it opens in a new tab.
My design disappeared	Autosave is per-browser. Export JSON for safe backups.
An item is highlighted red	It overlaps another item or sits outside all rooms.